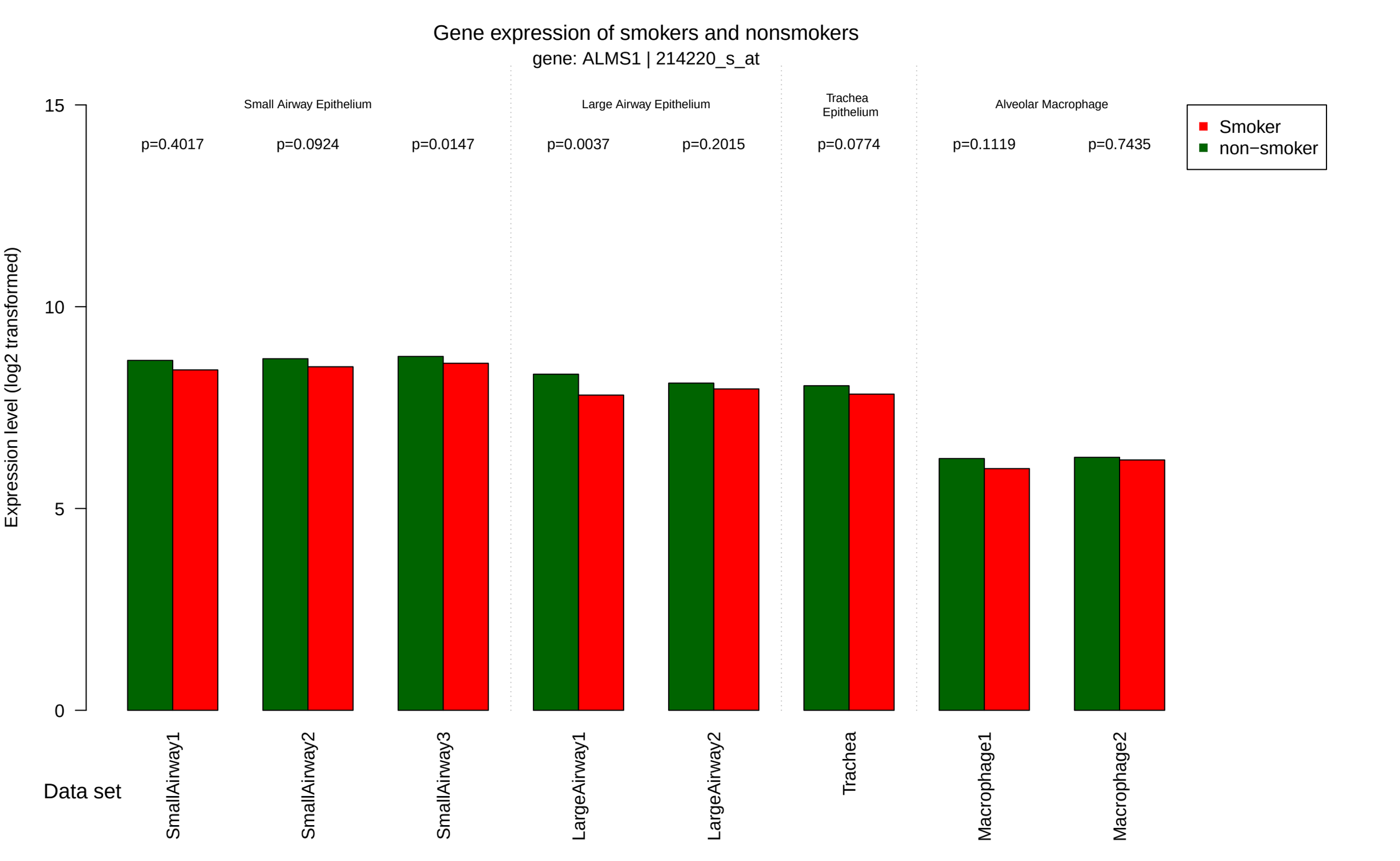
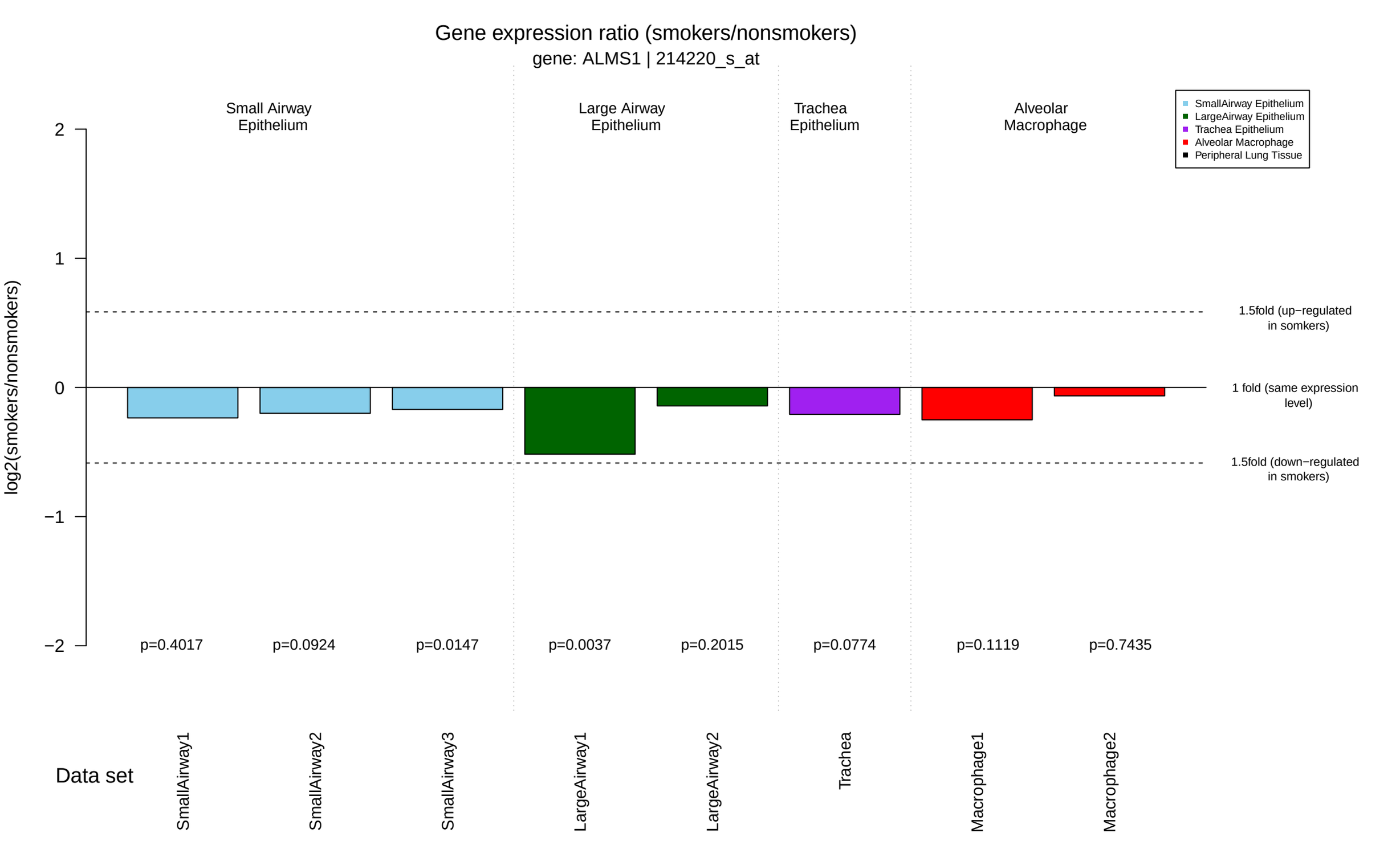
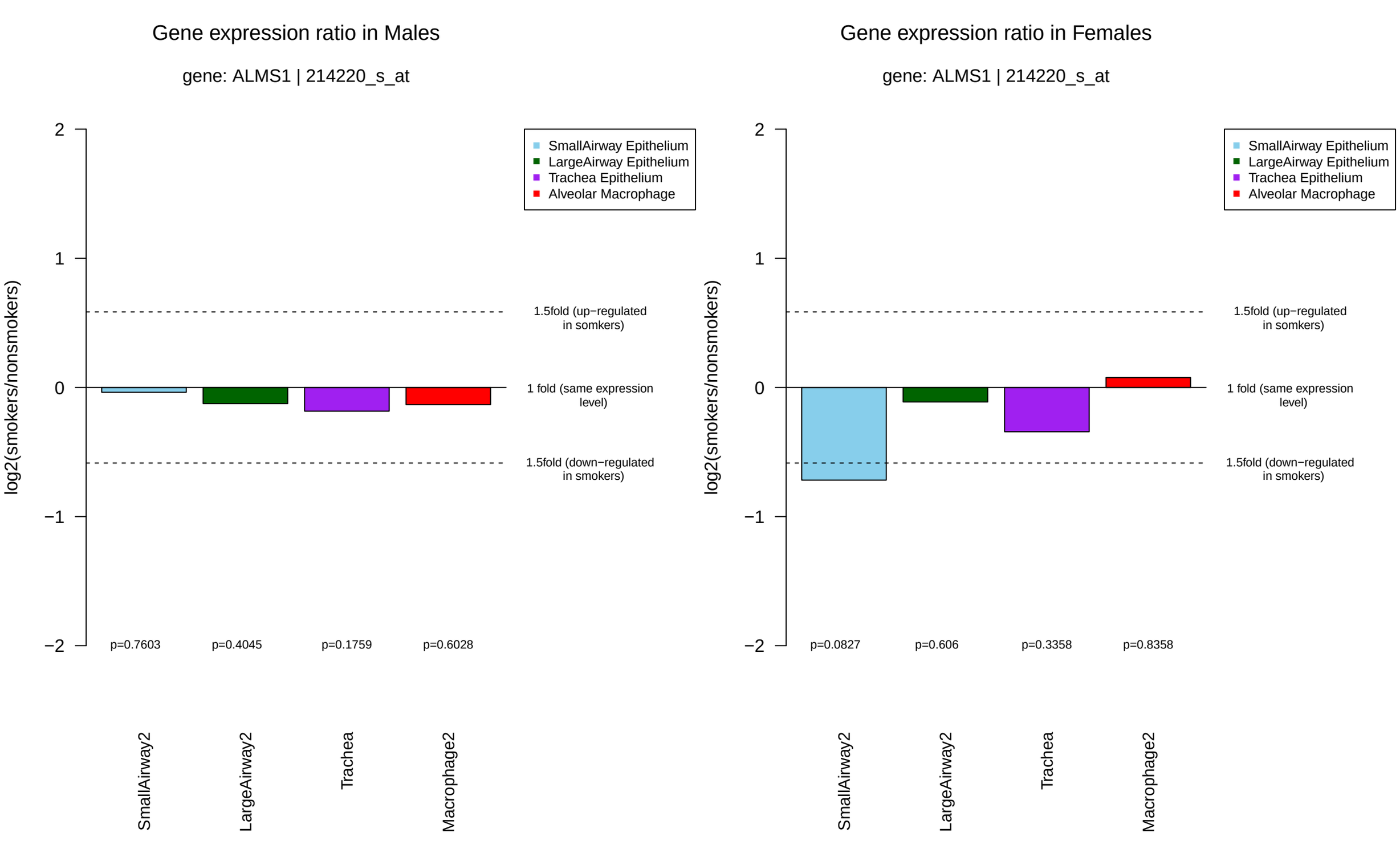
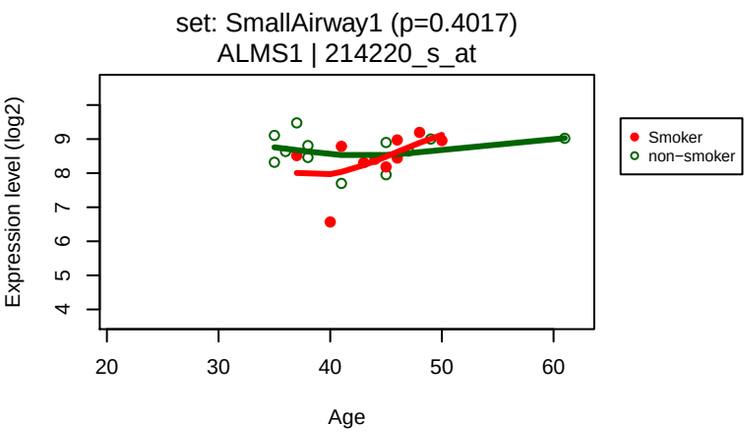
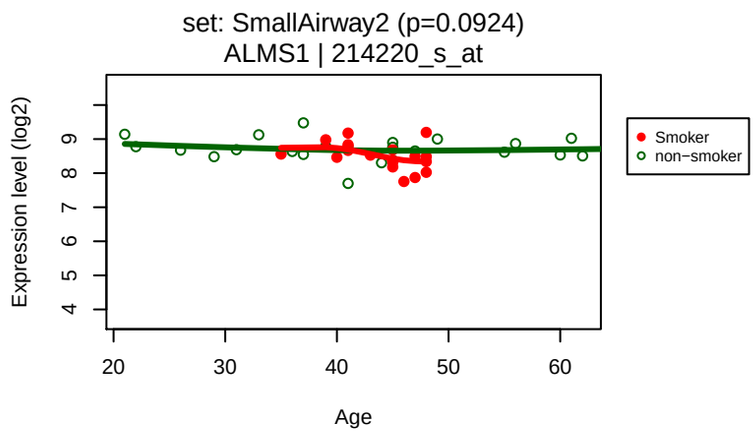
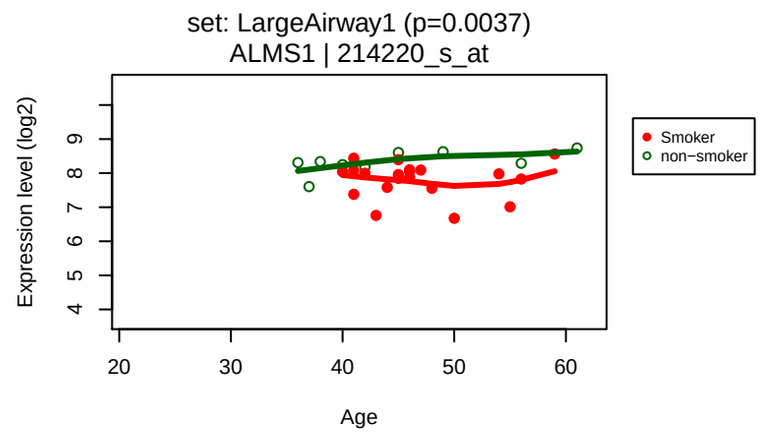

Gene expression of smokers and nonsmokers
gene: ALMS1 | 214220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4017
p=0.0924
p=0.0147
p=0.0037
p=0.2015
p=0.0774
p=0.1119
p=0.7435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

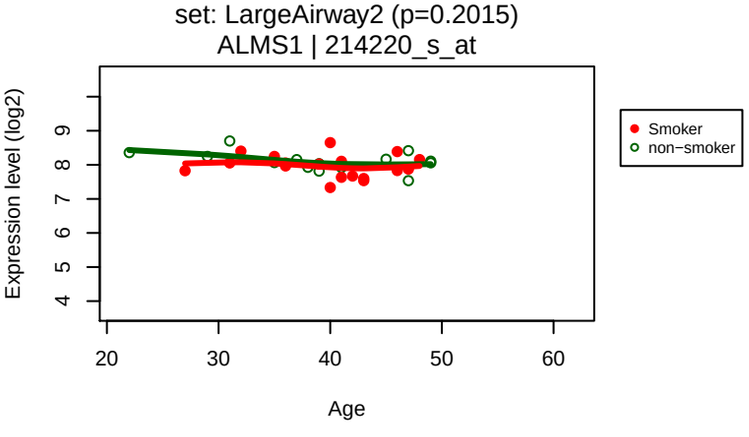

Gene expression ratio (smokers/nonsmokers)
gene: ALMS1 | 214220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4017
p=0.0924
p=0.0147
p=0.0037
p=0.2015
p=0.0774
p=0.1119
p=0.7435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

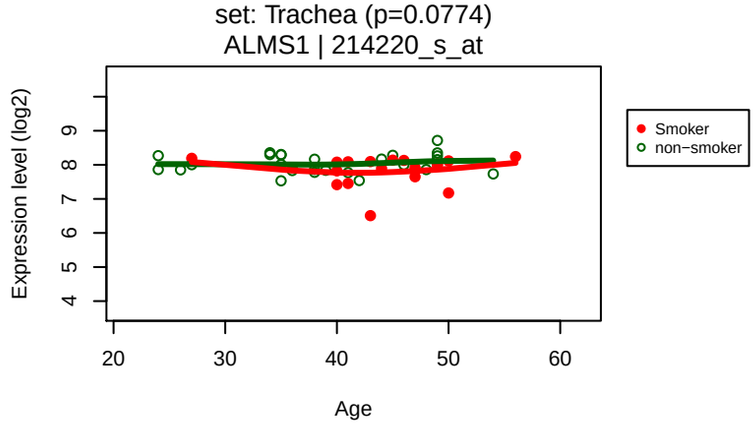

Gene expression ratio in Males
gene: ALMS1 | 214220_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7603
p=0.4045
p=0.1759
p=0.6028
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALMS1 | 214220_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0827
p=0.606
p=0.3358
p=0.8358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

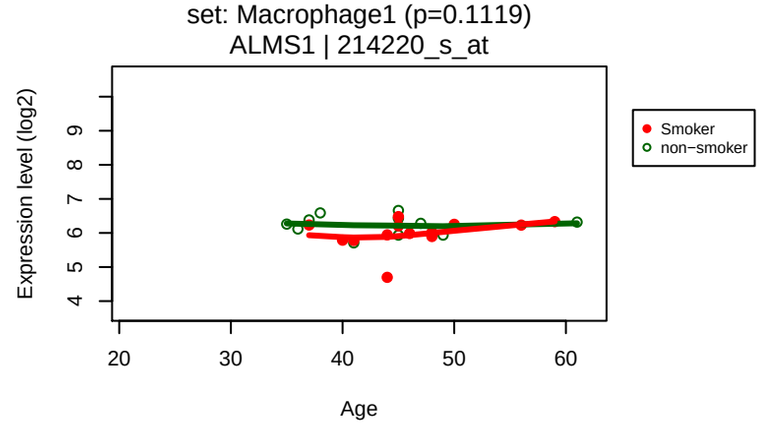

set: SmallAirway1 (p=0.4017)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker

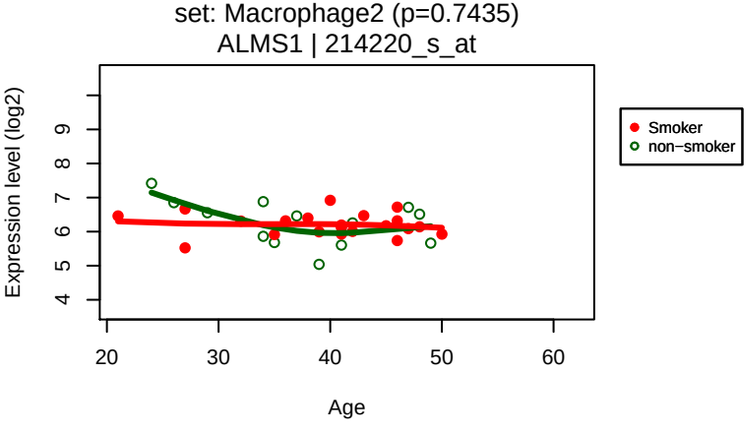

set: SmallAirway2 (p=0.0924)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker

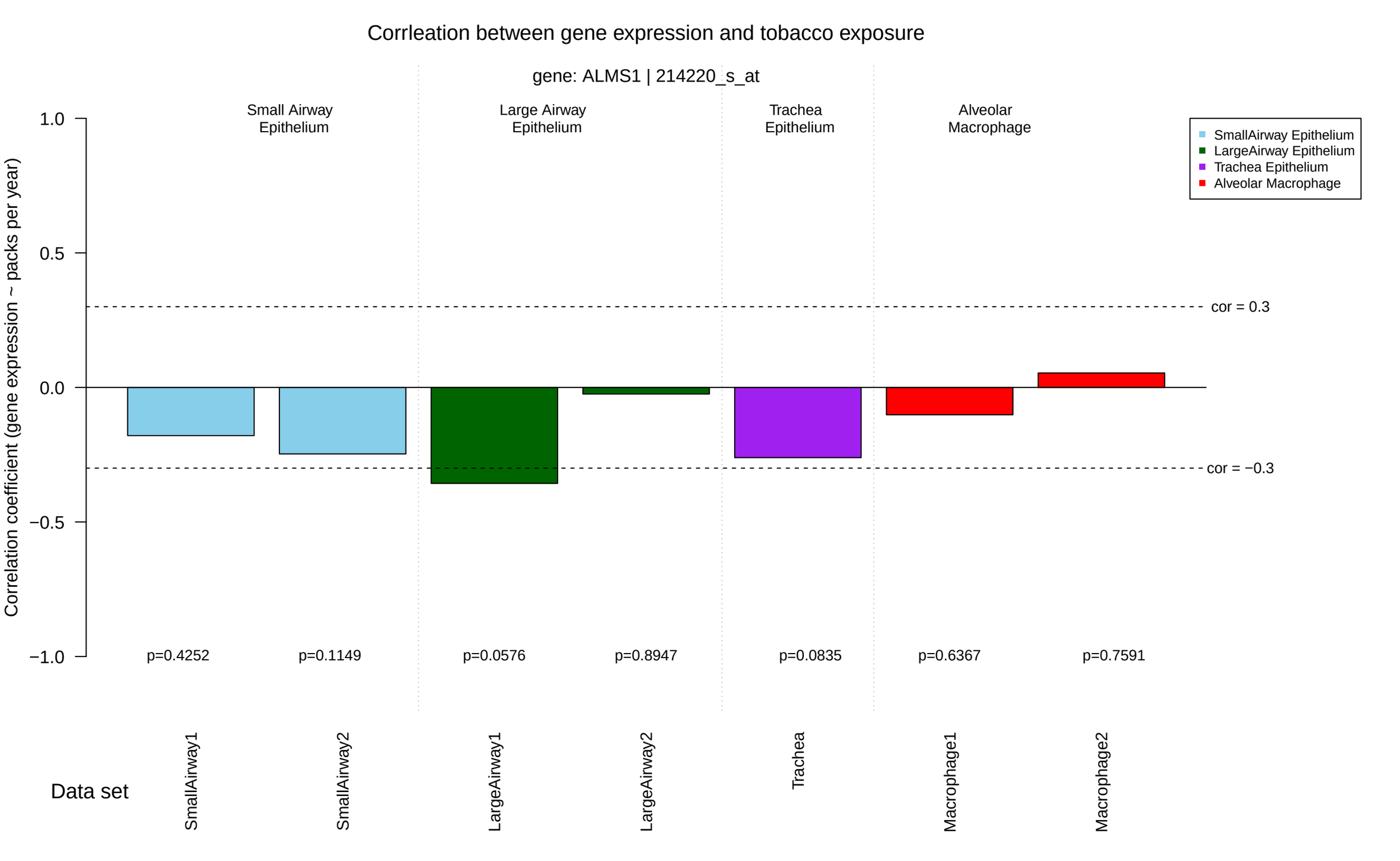

set: LargeAirway1 (p=0.0037)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2015)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0774)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1119)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7435)
ALMS1 | 214220_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALMS1 | 214220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4252
p=0.1149
p=0.0576
p=0.8947
p=0.0835
p=0.6367
p=0.7591
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2